Design IA (Indoor Chair)

By Arill Shukla

A1 Word Count = 234

Outlining the problem: In the modern era, one of the largest most widespread underlying issues is stress. Notably impacting younger individuals, where such issues are often the ramifications of societal pressure, expectations, relationships and finance. Data from crossrivertherapy.com suggests that 45% of students in high school and college face stress on a daily basis whereas referring to articles by BC (Best colleges), these end up having various impacts on their mental and physical wellbeing. These issues are then further escalated after students are made to sit in stiff classroom chairs which forces them to take awkward postures.

Design Situation: An environment where teenagers are seen using conventional uncomfortable seating. The students face anxiety and stress due to various reasons and thus are unhappy and unable to focus.

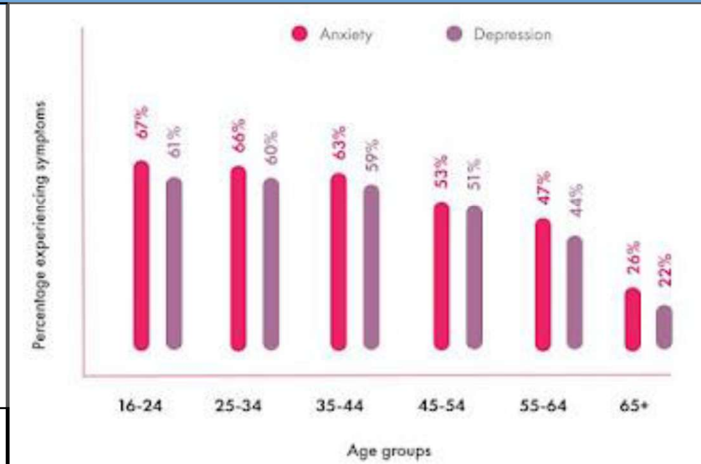


Figure 1: Age groups experiencing stress (Champion Health) The Workplace Health Report: 2023. (Joe P., 2023)

Situation: Teenagers needing a safe space. (Lounging area where they can relax and enjoy themselves)

-Stressed students
-Collective pressure on teenagers
-Inability to focus


Fig 11

User:

- Teenager from India who deals with stress daily.
- The student is anxious, unhappy and is unable to focus.
- Experiences headaches & low morale.
- Experiencing lower body back pain due to poor posture.
- Requesting a safe space where the Student can relax and enjoy themself.

User Ranges from 13-18 years of age.

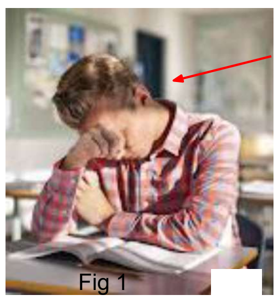


Fig 1

Stressed and Anxious

Awkward Slouched Posture

Trouble Focusing

Unhappy and Moody


Fig 2

Chair: Ikea GRÅSALA

Stiff Posture

Conventional Uncomfortable Seating

Existing product analysis: IKEA MALSKÄR Chair (Polypropylene plastic & Steel)


Fig 4


Fig 5

Lower back support but promotes arched back.
Height Adjustability (User
Minimalistic Design (Simple)

Basic Structure & Conventional swivel chair
Not personalized for stressed teenagers
Lacking in enjoyable factors & playfulness
Lacking in cushioning & comfort

No movement, rigid chair structure

Hard surface (no pressure support)

Figure 3: SDG 3 & SDG 11 (Sustainable Development Goals)


Fig 6

Enjoyment & Functionality Factor
Rotating & Moving wheels to help with mobility. (Swivel)

Figure 8: Common Symptoms of Stress. (NYCHealthy, 2022)

Safe Spaces for Teens Aren't Controversial. They're Critical. Here's Why.

When the researchers Carl Hanson and Quinn Snell set out to identify the top 10 factors that predicted suicidal thoughts and behavior in 179,000 Utah high school students, they had no preconceived notions. Instead, they fed years worth of survey responses from those teens, who'd answered questions about things like school involvement, family life, and mental health, into 100 different machine learning models, eager to let the data lead them to a conclusio

Figure 9: Article on the importance of Safe Spaces for Teens (CHC, Rebecca R.)

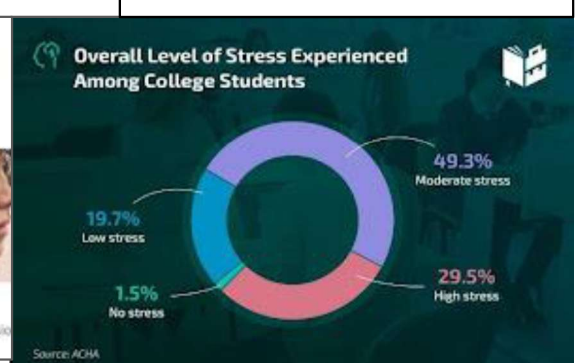
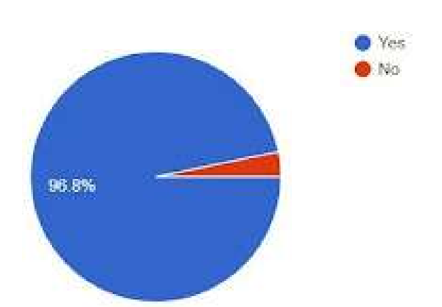


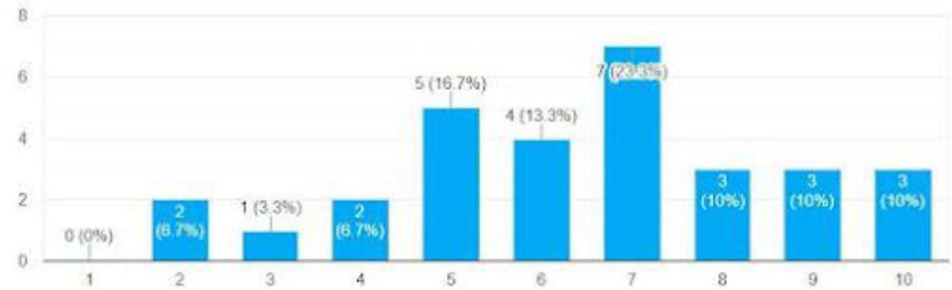
Figure 10: Pie chart on stress levels among college students (Whattobecome, 2022)

Survey for Data Collection (31 responses)

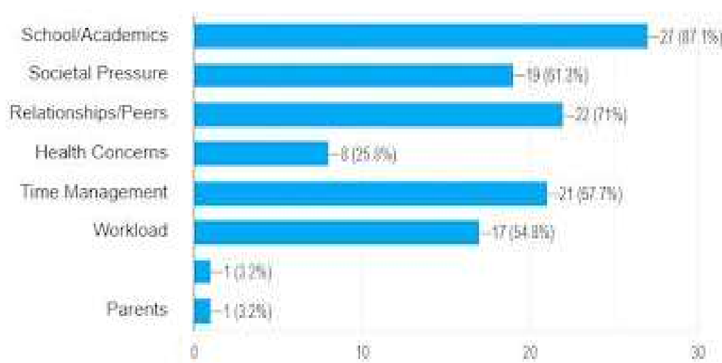
As a teenager, do you feel stressed?



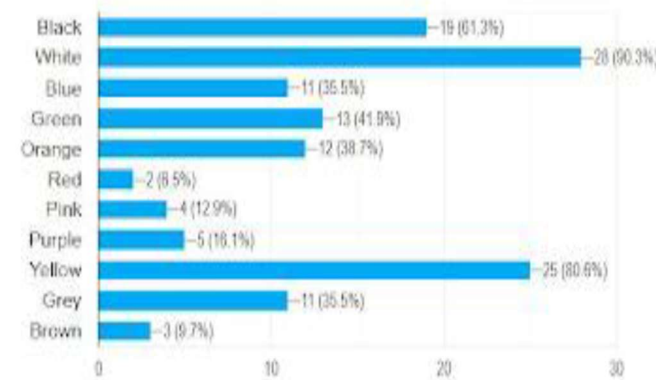
If you do feel stressed, how often?



What are some reasons for your stress?



What are some colours that you find soothing?



What are some of the impacts of stress on you?

I often have a hard time studying and I see myself getting angrier much more often.

The quality of work I do depreciates and I barely have energy.

I get depressed and ,my head hurts sometimes

Less sleep, can't focus, feel sleepy all the time

When I have lots of things on my mind, I end up doing none of them and that makes the stress even worse.

spiraling and pessimism

Depression, Anxiety, Sleep deprivation

I always feel like I'm doing nothing and that I'm worse than others because there's so many different things going on that I don't think I can handle.

Photos of Environment (Student Lounge)


Fig 13

Soothing colours

Cozy atmosphere

Comfortable & Cozy Seating

Plenty of space to enjoy.

Fun shapes and enjoyable ambiance.

Target User: Teenagers from 13-18 who require a safe space due to the collective stress of their lifestyle

Fig 14

Values of IKEA

- Togetherness
- Caring for people and planet
- Cost consciousness
- Simplicity
- Renew and improve
- Different with a meaning
- Give and Take responsibility
- Lead by example

Creating a product for IKEA means creating a product which is high in quality functional, affordable, and sustainable at the same time- As Ikea is committed to taking responsibility for its environmental impacts. For the design, the product would preferably emphasize simplicity, functionality, and clean lines. Lastly, the product must have a flat-pack design which allows for easy transportation which greatly reduces shipping costs, environmental impact, and allows customers to enjoy the experience of putting the product together. (Further decreasing costs)

CRITERIA A2: DESIGN BRIEF

A2 Word Count = 175

Fig 15

Spun (2023) *Magis S.p.A.*(Accessed: 19 September 2023)

Fig 16

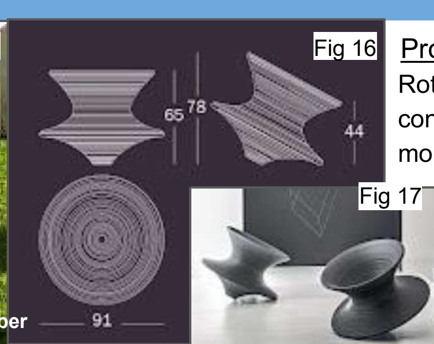


Fig 17

Product: Spun - Magis

Rotating Chair- moves primarily due to the conservation of angular momentum (product of the moment of inertia and its angular velocity.

Enjoyable and entertaining due to rocking

Unconventional (form and angular rotation)

Simple and minimalistic

Lacks padding (Stiff and rigid

Lacks proper backrest

Does not utilize flatpack

My aim is to: design a one-off high quality (high fidelity) chair prototype that serves as lounge seating for stressed teenagers. This product must reduce tension in the back, fatigue, stress, and other psychological issues. Furthermore preventing the possible repercussions of a distressed mood.

The target user: The primary target users are teenagers (aged 13-18) who may be negatively affected by the cumulative stresses of academics, relationships, societal expectations and many others. Whereas my research suggests that teenagers and students would be most interested in my presented solution.

The product must: Reduce distress teenagers to prevent any repercussions. The chair must provide as an enjoyable, playful, and comfortable seating which allows these teenagers to relax and let loose. It must be unconventional, interactive and engaging where it should be fun to sit in. Lastly, the chair must take on a soothing aesthetic which maintains the students peace of mind and must stay stable and safe to avoid any injuries.

CRITERIA A3: SPECIFICATIONS

A3 Word Count = 1038

Specific Category	Specific Specification Point	Justification
Function 1.0	1.1: Product must be include movement such as rotation to make the product more engaging and interactive.	The chair should offer comfort, engage users with interactive features, and have a soothing aesthetic. By reducing emotional stress, the goal is to mitigate the adverse effects of stress on their mental and physical well-being sustainably. (Manufacturing)
	1.2: Product must invoke feelings of joy and relax distressed teenagers.	Designing the chair to invoke joy and relaxation is crucial as it directly addresses the emotional needs of distressed teenagers. Creating a positive and soothing experience through user-centered design (UCD) helps alleviate stress, fostering a sense of well-being and emotional comfort. (User Feedback)
	1.3: Product must be able to accompany 95th percentile of weight (35 BMI).	Ensuring the chair accommodates individuals with a 35 BMI (95th percentile of weight) is essential for inclusivity and safety. It prevents potential structural issues and guarantees that a broad range of users, including those with varying body types, can benefit from the chair's stress-relieving features comfortably. (User Trial)
	1.4: Product must be comfortable & provide cushioning.	Ensuring comfort and cushioning is vital to promote relaxation and reduce stress effectively. It enhances the overall user experience, making the chair a welcoming and cozy space for distressed teenagers to unwind, fostering emotional well-being. (Manufacturing)
	1.6: Product must be accessible.	Ensuring that people in all areas of the world and those with issues in mobility, disabilities are able to purchase and use the chair comfortably. Being accessible would furthermore allow inclusivity promoting diversity. (Test Product Launch)
	1.5: Product must be able to assemble and disassemble (Flatpack)	Allowing the chair to be easily assembled and disassembled facilitates practicality and versatility. Simplifying transportation, storage, potential adjustments, and fulfills IKEA's requirements for a flat pack product. This also further reduces economic expenses. (Manufacturing)
Size/ Dimensions 2.0	2.1: The product must cater to a waist circumference of 119 cm (95th percentile).	By utilizing anthropometric data, the chair is able to accommodate a waist circumference of 119 cm (95th percentile) ensuring inclusivity for users with diverse body sizes. This aligns with the principles of fairness and user-centered design. (stacks.cdc.gov) (Conduct User Trial)

Specific Category	Specific Specification Point	Justification
Sustainability 3.0	3.2: Product must be easy to deconstruct at the end of the product life cycle.	Enabling easy deconstruction promotes sustainability by facilitating recycling and responsible disposal, reducing environmental impact at the end of the chair's life cycle. (User trial)
Target Audience 4.0	4.1: Product must be able to appeal to teenagers and students.	Designing with teenage appeal ensures the chair effectively addresses the needs and preferences of the target demographic, enhancing its impact. (User Feedback)
	4.2: Product must reduce anxiety and distract the target audience.	The chair's ability to reduce anxiety and distractions directly contributes to the well-being of teenagers, promoting emotional health and focus on academic tasks. (User Feedback)
Product Constraints 5.0	5.1: Prototype must use resources provided in school workshop.	Using school workshop resources minimizes costs, fosters resourcefulness, and promotes practical learning in a controlled educational environment. (Manufacturing)
	5.2: Prototype must use tools provided in school workshop.	Utilizing school workshop tools ensures accessibility and familiarity, simplifying the construction process and enhancing the feasibility of the project. (Manufacturing)
Aesthetics 6.0	6.1: Product must look unconventionally playful and fun.	An unconventional, playful, and fun appearance aligns with the emotional well-being goal, making the chair more appealing and engaging for teenagers, thereby increasing its effectiveness. (User Feedback)
	6.2: Product must use soothing colours.	Incorporating soothing colors enhances the chair's calming effect, creating a peaceful atmosphere for distressed teenagers, which is vital for reducing anxiety and stress. (User Feedback)
Safety 7.0	7.1: Product must be high quality and avoid breaking down randomly.	Ensuring the chair avoids random breakdowns guarantees its reliability and longevity, providing consistent support to users and maintaining their trust. (Product Testing)
	7.2: Product must handle a weight of 250 kilos (113 kg's)	Being able to handle weight is essential for a chair for prolonged periods of time, making sure the chair does not break down. The 95th percentile of weight in 18 year olds is 103 kg's (lifemeasure.com), whereas the standard chair handles 113 kg's.(conceptseating.com) (User Trial)
	7.3: Product must have beveled edges	Smooth edges are essential for safety to prevent injuries and improve user comfort. (Manufacturing)
Material Selection 8.0	8.1: Product must have comfortable cushioning. (eco-friendly)	Comfort is essential for relaxation and well-being, making comfortable cushioning a necessary feature to achieve the chair's stress-reducing goals effectively. (Manufacturing)
	8.2: Product must use material resistant to abrasive surfaces, repeated use, and be durable. (eco-friendly)	Resistance to abrasion and repeated use is crucial for the chair's durability and safety, ensuring it withstands the wear and tear associated with active teenagers. The durability of the chair would furthermore ensure stability in the chair. (User Trial)
	8.3: Product must use soft and smooth materials. (eco-friendly)	Soft and smooth materials enhance the tactile experience, contributing to comfort and the overall appeal of the chair, which is important for attracting and retaining teenage users. (Manufacturing)

Target User 9.0	9.1: Product must reach teenagers and students around the age of 13-18.	Focusing on the 13-18 age group ensures the chair addresses the specific needs and preferences of its target audience, optimizing its effectiveness in reducing stress. Moreso, teenagers require comfortable and ergonomic chairs to help support their constantly growing bodies. This makes them a vital demographic.
Lifespan 10.0	10.1: Product must be high quality to maintain itself for as long as possible.	High-quality construction is essential to maintain the chair's functionality and aesthetics over time, ensuring it serves its stress-reduction purpose effectively for as long as possible. (User Trial)
Quantity 11.0	11.1: Product must have one CAD, One Low Fidelity, and one High fidelity prototype. Portraying Aesthetics and Functionality.	Combining functionality and aesthetics in the prototype demonstrates the chair's potential for both practical use and visual appeal, making it more appealing to users and potential stakeholders. (Manufacturing)
Cost 12.0	12.1: Product must cost around 6,000 rupees.	High quality IKEA chairs are around the price range of 6,000 rupees, this budget allows for the product to be made of high quality materials whilst offering comfort, durability, and various ergonomic features without being affordable by a range of consumers.(Test Product Launch)

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A1 Word Count = 290

Outlining the problem: Park benches, one of the most common chairs we come across. The conventional design of this seating usually follows an uncomfortably rigid structure on which people are usually unable to sit on for long periods of time. Not to mention, these park benches lack various features such as a shade, power plugs for charging, and something to prevent people from sleeping. The chairs should be designed in a much more comfortable way while still maintaining its durability, possibly paired with a side table to keep any belongings.


CISION
News Products Contact

May's Posture Awareness Month Educates and Combats the Negative Health Complications of Bad Postures

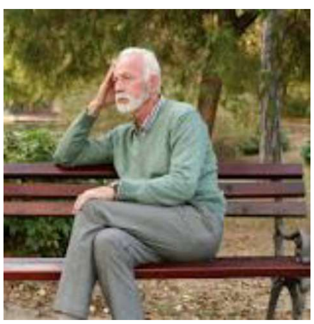
Posture Awareness Month's campaign theme, "Stand Taller for Posture Month", brings together thousands of doctors, therapists and trainers in response to the epidemic health consequences of poor posture related to modern technology.

Figure 1: Age groups experiencing stress (Champion Health) The Workplace Health Report: 2023. (Joe P., 2023)

Design Situation: A park with people sitting and sleeping in uncomfortable and rigid benches. Sunshine directly hits these benches irritating those sitting on these seats, those whose electronic devices run out of battery are unable to charge from anywhere and have nowhere to keep their belongings except on the bench, occupying an extra seat.

Figure 2: SDG 3 & SDG 11 (Sustainable Development Goals)

User:
- 46 year old Australian man who enjoys long walks.
- Requires momentary breaks due to fatigue and old age.
- Requires a bench to keep belongings and a charging port to charge electronics.
- Sits in awkwardly uncomfortable positions due to the design of benches, often affecting his posture.
- Irritated due sleeping people occupying benches and sunshine, giving headaches and ruining his mood
- Requesting a bench that solves all these issues.



Awkward posture

Irritation and stress

Taking break from long walk.

Glaring Sunshine

Uncomfortably rigid Bench

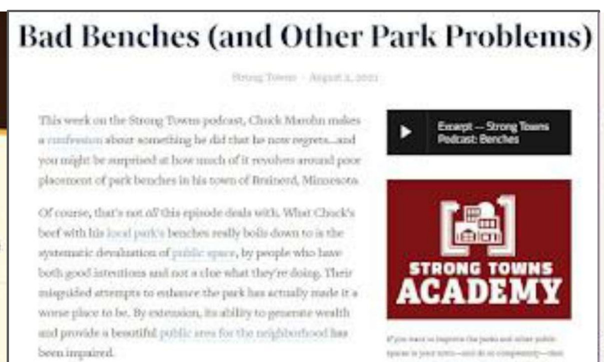
User Ranges from 18-60 years of age.

Existing product analysis

-Back support at good angle
-Insulating material (Wood)
-Simple and minimalistic

-Lacks charging ports
-Lacks shade
-Allows people to sleep
-Rigid and stiff
- Lacks side table

Figure 3: Prnewswire.com (Bodynews 2016)


Bad Benches (and Other Park Problems)

This week on the Strong Towns podcast, Chuck Marohn makes a confession about something he did that he now regrets...and you might be surprised at how much of it revolves around poor placement of park benches in his town of Brainerd, Minnesota.

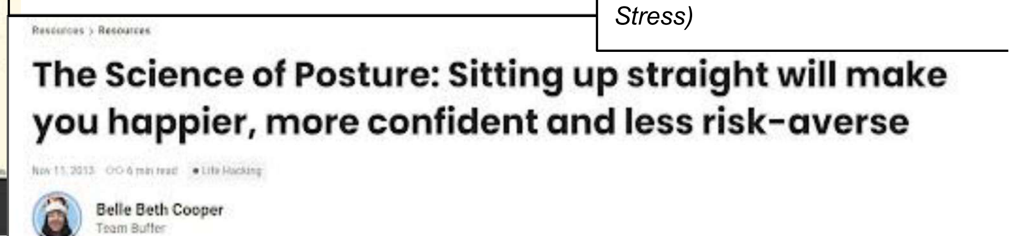
Of course, that's not all this episode deals with. What Chuck's beef with his local park's benches really boils down to is the systematic devaluation of public space, by people who have both good intentions and not a clue what they're doing. Their misguided attempts to enhance the park has actually made it a worse place to be. By extension, its ability to generate wealth and provide a beautiful public area for the neighborhood has been impaired.

Excerpt — Strong Towns Podcast: Benches

STRONG TOWNS ACADEMY

Figure 5: Strong town (Bad benches, 2021)

Figure 6: APA.org (Generation Stress)


Resources › Resources

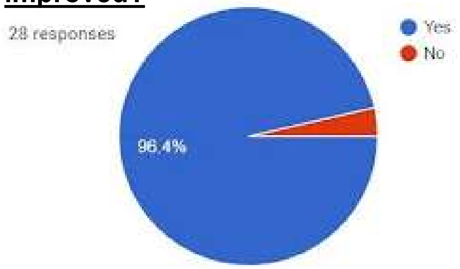
The Science of Posture: Sitting up straight will make you happier, more confident and less risk-averse

Belle Beth Cooper
Team Buffer

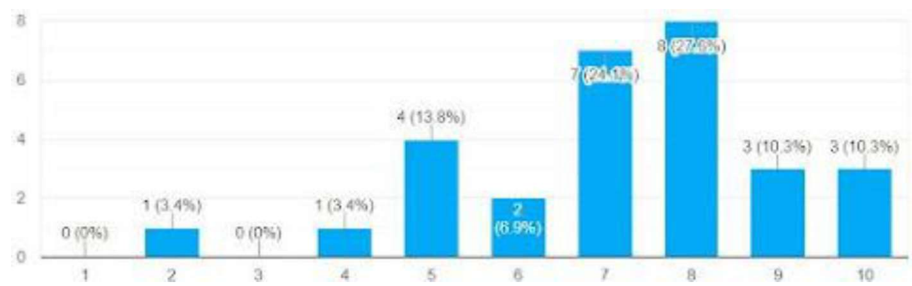
Figure 4: Buffer.com (Belle beth Cooper)

Survey for Data Collection (29 responses)

Do you think park benches should be improved?



How often do you go out for walks?



What are some common issues in park benches?

I often forget to charge my electronics, so I think a charging point would be useful

i cant go on a bench in the morning due to the sunlight

Back pain

people sleeping on the benches!

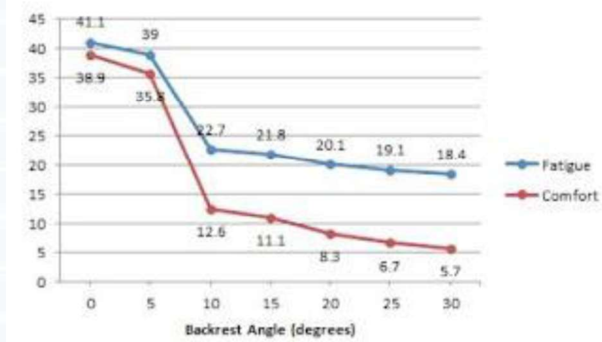
uncomfortable arm and backrests

Too much rust and decay from the years

lack of maintainence so not too safe

non eco-friendly use of materials.

Optimal angle for fatigue and comfort



https://www.jstage.jst.go.jp/article/ijat/10/2/10_153/_pdf (2016)

(Than L., Tharindu D., Vishan A., Vinura W.)

How do these issues affect you?

lower back pain and poor posture

I don't use the benches because of all the sunlight

unable to find a sitting spot due to people sleeping

I can't charge my devices

due to the low maintenance, the chair risks falling apart due to rust

sitting in awkward positions irritates me

Photos of Environment (Park)

Lots of greenery

Glaring Sunlight

Public space (Large number of people)

People Sleeping on benches

Passerby on walks

Target User: Age 18-60 who require more comfortable, efficient, and improved park benches.

Values of IKEA

- Togetherness
- Caring for people and planet
- Cost consciousness
- Simplicity
- Renew and improve
- Different with a meaning
- Give and Take responsibility
- Lead by example

Creating a product for IKEA means creating a product which is high in quality functional, affordable, and sustainable at the same time- As Ikea is committed to taking responsibility for its environmental impacts. For the design, the product would preferably emphasize simplicity, functionality, and clean lines. Lastly, the product must have a flat-pack design which allows for easy transportation which greatly reduces shipping costs, environmental impact, and allows customers to enjoy the experience of putting the product together. (Further decreasing costs)

CRITERIA A2: DESIGN BRIEF

A2 Word Count = 160

Product: Korkat (Shaded Bench)

KorKat park bench provides the comfort of shade on an 8 foot bench.

Shaded to block glaring sunlight
Simple design
Proper angled backrest
Stiff and rigid
Does not utilize flatpack
Lacks charging point to charge electronics
Lacks side table

My aim is to: design a one-off high quality (high fidelity) seating area which is both efficient and effective when serving as a park bench. This product must reduce tension in the back and fatigue while being equipped with various features which would improve user experience.

The target user: The primary target users are teenagers to elderly (aged 18-60) who face various issues when trying to use a park bench. This includes those who are unable to find a bench which is occupied by someone sleeping, a bench without shade, a bench without a charging port etc.

The product must: be comfortable for the user and utilize all features efficiently. The bench must have an adjustable shade to block sunlight, charging ports, be high quality to not fall apart, have multiple armrests (hostile architecture) to ensure people would not sleep on it, and be sustainable.

CRITERIA A3: SPECIFICATIONS

A3 Word Count = 1100

Specific Category	Specific Specification Point	Justification
Function 1.0	1.1: Product must be equipped with barriers and multiple armrests	The chair should offer armrests and barriers in the middle of its body to prevent people from sleeping. This would solve the issue of insufficient benches and allow more people to use them. (User Trial)
	1.2: Product must have charging points	The bench must be equipped with charging ports on either side allowing users to charge their devices whenever required. To be sustainable, these must be solar powered and would allow the user to save much more time.
	1.3: Product must be able to accompany 95th percentile of weight (35 BMI).	Ensuring the chair accommodates individuals with a 35 BMI (95th percentile of weight) is essential for inclusivity and safety. It prevents potential structural issues and guarantees that a broad range of users. (User Trial)
	1.4: Product must have an adjustable shade and provide comfort.	Ensuring comfort is simply vital to promote relaxation and reduce stress effectively. It enhances the overall user experience, making it easier for the user to unwind. By ensuring the angle of the bench backrest is optimized, pressure of the backbone would be taken off. (Manufacturing)
	1.5: Product must be accessible.	Ensuring that people in all areas of the world and those with issues in mobility, disabilities are able to purchase and use the chair comfortably. Being accessible would furthermore allow inclusivity promoting diversity. (Test Product Launch)
	1.6: Product must be able to assemble and disassemble (Flatpack)	Allowing the chair to be easily assembled and disassembled facilitates practicality and versatility. Simplifying transportation, storage, potential adjustments, and fulfills IKEA's requirements for a flat pack product. This also further reduces economic expenses. (Manufacturing)
	1.7: Product must have an adjustable shade.	Ensuring that the bench would have an adjustable shade would allow users to come by at any time and use the shade to relax and block themselves from the sunlight and heat anytime they want. By making it adjustable, users would be able to modify it to their liking. (Manufacturing)

Specific Category	Specific Specification Point	Justification
Size/ Dimensions 2.0	2.1: The product must cater to a waist circumference of 119 cm (95th percentile).	By utilizing anthropometric data, the chair is able to accommodate a waist circumference of 119 cm (95th percentile) ensuring inclusivity for users with diverse body sizes. This aligns with the principles of fairness and user-centered design. (stacks.cdc.gov) (Conduct User Trial)
Sustainability 3.0	3.1: Product must be easy to deconstruct at the end of the product life cycle.	Enabling easy deconstruction promotes sustainability by facilitating recycling and responsible disposal, reducing environmental impact at the end of the chair's life cycle. (User trial)
Target Audience 4.0	4.1: Product must be able to appeal teenagers and elderly.	Designing with target audience appeal ensures the chair effectively addresses the needs and preferences of the target demographic, enhancing its impact. (User Feedback)
	4.2: Product must reduce stress and allow users to unwind.	The chair's ability to reduce anxiety and allows users to unwind contributes to the comfort of the bench, further optimizing the users experience. (User Feedback)
Product Constraints 5.0	5.1: Prototype must use resources provided in school workshop.	Using school workshop resources minimizes costs, fosters resourcefulness, and promotes practical learning in a controlled educational environment. (Manufacturing)
	5.2: Prototype must use tools provided in school workshop.	Utilizing school workshop tools ensures accessibility and familiarity, simplifying the construction process and enhancing the feasibility of the project. (Manufacturing)
Aesthetics 6.0	6.1: Product must look simple and cozy.	By increasing the chairs aesthetic appeal, the product would attract many more passerby thus fulfilling its role and bringing satisfaction to many users. (User Feedback)
	6.2: Product must use soothing colours.	Incorporating soothing colors enhances the chair's calming effect, creating a peaceful atmosphere. (User Feedback)
Safety 7.0	7.1: Product must be high quality and avoid breaking down randomly.	Ensuring the chair avoids random breakdowns guarantees its reliability and longevity, providing consistent support to users and maintaining their trust. (Product Testing)
	7.2: Product must handle a weight of 250 kilos (113 kg's)	Being able to handle weight is essential for a chair for prolonged periods of time, making sure the chair does not break down. Whereas the standard chair handles 113 kg's.(conceptseating.com) (User Trial)
	7.3: Product must have beveled edges	Smooth edges are essential for safety to prevent injuries and improve user comfort. (Manufacturing)
Material Selection 8.0	8.1: Product must have comfortable cushioning. (eco-friendly)	Comfort is essential for relaxation and well-being, making comfortable cushioning a necessary feature to achieve the chair's stress-reducing goals effectively. (Manufacturing)
	8.2: Product must use material resistant to abrasive surfaces, repeated use, and be durable. (eco-friendly)	Resistance to abrasion and repeated use is crucial for the chair's durability and safety, ensuring it withstands the wear and tear of the everyday public. The durability of the chair would furthermore ensure stability in the chair. (User Trial)

Material Selection 8.0	8.3: Product must use soft and smooth materials. (eco-friendly)	Soft and smooth materials enhance the tactile experience, contributing to comfort and the overall appeal of the chair, which is important for attracting and retaining passerby. (Manufacturing)
Target User 9.0	9.1: Product must reach teenagers and elderly around the age of 18-60.	Focusing on a large age group makes it tougher to address specific needs and preferences of its target audience, but both teenagers and elderly require comfortable and ergonomic chairs to help support their constantly growing bodies. This makes them a vital demographic.
Lifespan 10.0	10.1: Product must be high quality to maintain itself for as long as possible.	High-quality construction is essential to maintain the chair's functionality and aesthetics over time, ensuring it serves its purpose effectively for as long as possible. (User Trial)
Quantity 11.0	11.1: Product must have one CAD, One Low Fidelity, and one High fidelity prototype. Portraying Aesthetics and Functionality.	Combining functionality and aesthetics in the prototype demonstrates the chair's potential for both practical use and visual appeal, making it more appealing to users and potential stakeholders. (Manufacturing)
Cost 12.0	12.1: Product must cost around 15,000 rupees.	High quality IKEA chairs are around the price range of 6,000 rupees, however as the chair is a bench, a budget of 15,000 allows for the product to be made of high quality materials whilst offering comfort, durability, and various ergonomic features without being affordable by a range of consumers.(Test Product Launch)

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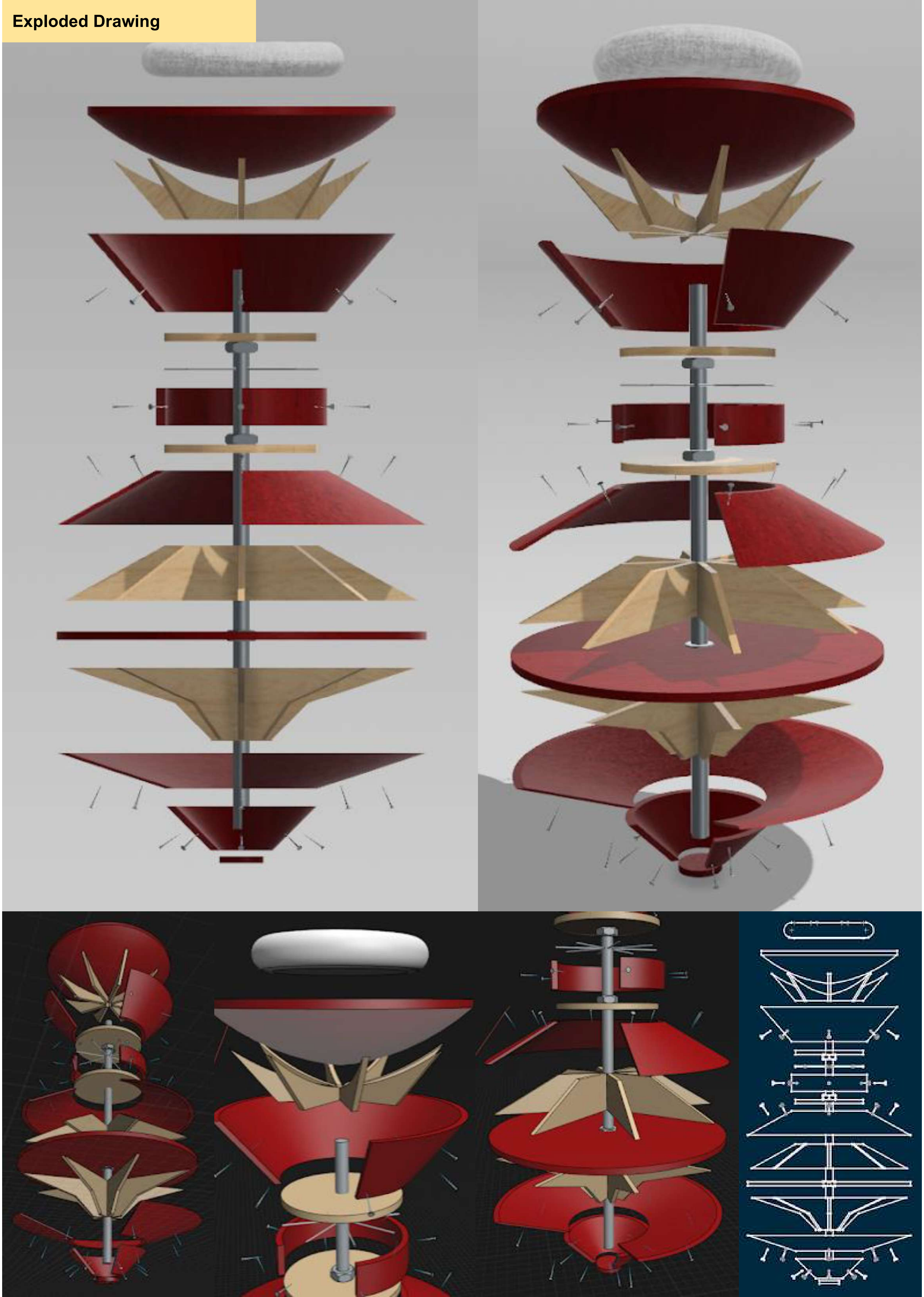
CRITERIA C1: THE PROBLEM

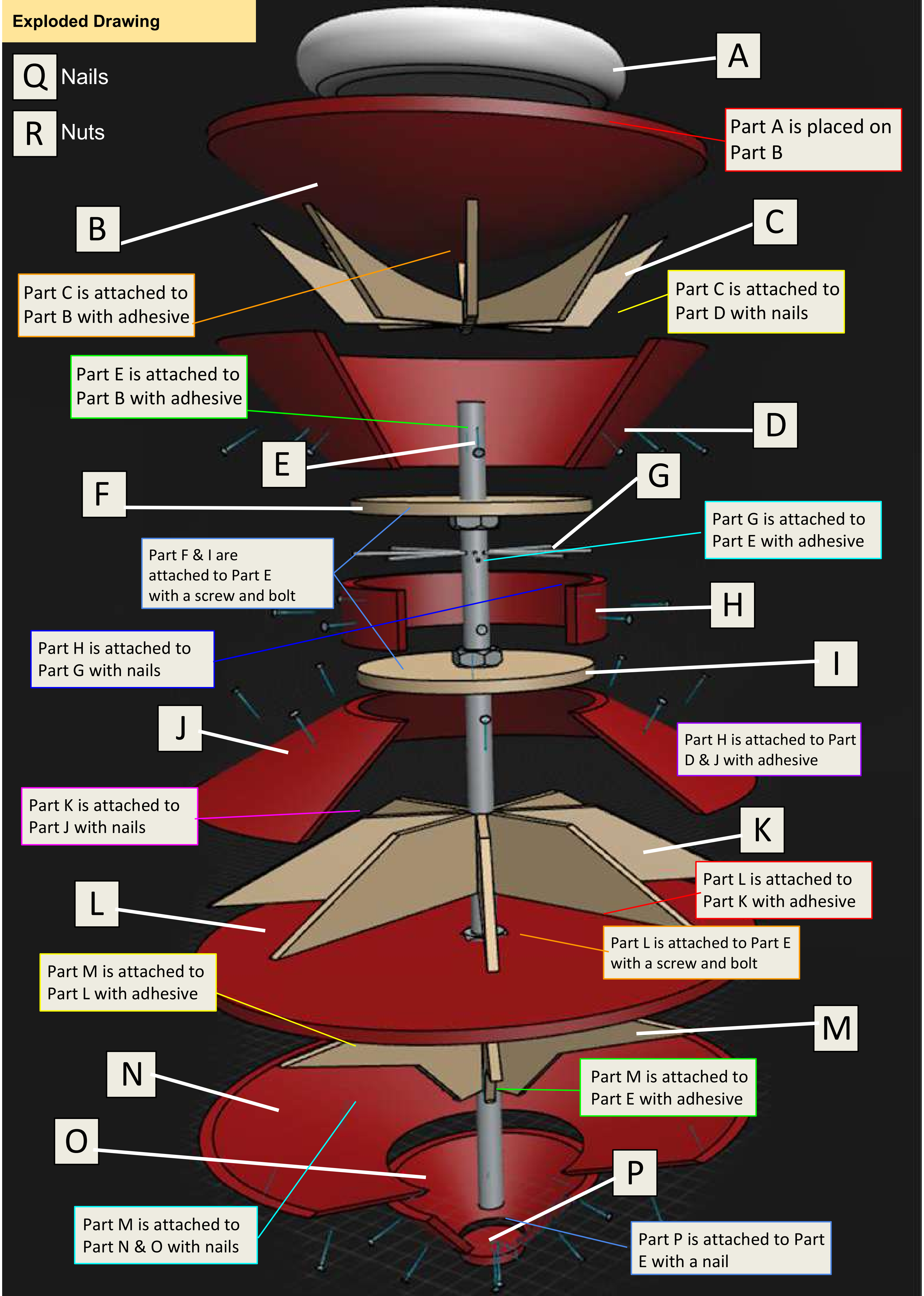
C1 Word Count = 297/ 400

No.	Components
1	Base Seat
2	Wooden Ribs
3	Iron Spokes
4	Iron Pole
5	Wooden exterior
6	Nuts
7	Bolts
8	Nails

Materials	Properties	Parts/ Components	Tools/ Machines
Pine wood	Pine is a softwood. thus giving it a hard, yet easy to model and carve, durable, dense and compact grained property. This is desirable due to its stability, ability to withstand weight, easy workability, and availability.	Oak is a suitable material for the outer exterior encasing of the chair, and to further support the beams inside the chair.	Tools: Chisel, Sand paper, Files, Clamps, File Machines: Table saw, Band saw, CNC router machine, Power drill, Orbital sander, Brushes (for coating)
Rubber	Rubber is flexible, elastic, Non-slip due its high coefficient of friction (Strong grip), Resistive to abrasion, and Shock absorbent.	Rubber is suitable to create the rubber feet/base of the chair to avoid abrasion, noise, and keep a strong centrepoint of rotation.	Tools: Rotary cutters/ Rotary knives, Files Machines: Injection/Compression molding, Die-cutting Machine, Laser cutting Machine, Extruding Machine,
Steel rod	Steel is known to be extremely strong (used for structure), durability, malleability, ductility and recyclability.	Steel is suitable for the steel rod and structural internal parts within the chair.	Tools: Angle grinders, Hacksaw, Power Drill Machines: Milling Machine, Drilling machine, Lathes, Belt sander, Grinder, Drill
Linen fabric	Linen is known for having a soft and smooth texture, Moisture wicking property, Durability, breathability, and Hypoallergenic properties (Resists dust and allergens)	Linen is suitable for the cushion used on the chair, stuffed with Polyester fiberfill.	Tools: Fabric scissors, Rotary cutter, Tailors chalk/Fabric marker measuring tape, seam gauge, Embroidery Hoops and Needles, Thread Machines: Sewing machine
Polyester fiberfill	Polyester fiberfill is known for being resilient to change in shape, Fluffiness, lightweight, easy maintenance and cost effectiveness.	Polyester fiberfill is suitable to stuff the linen to create a cushion on the chair.	Tools: Scissors, Fabric marker Machines: Sewing machine, Air compressor these Compressors have various uses(Industrial)

Exploded Drawing

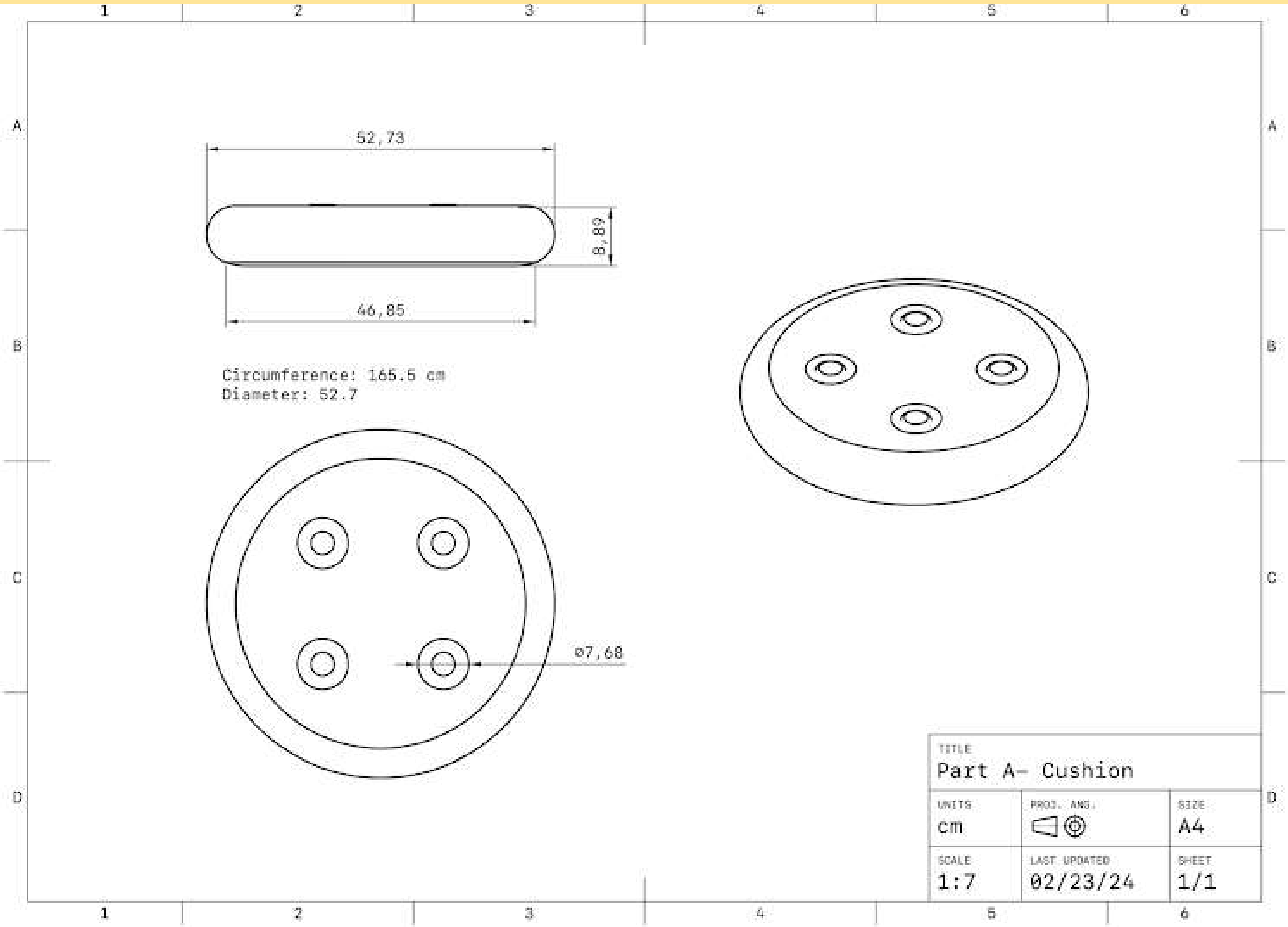


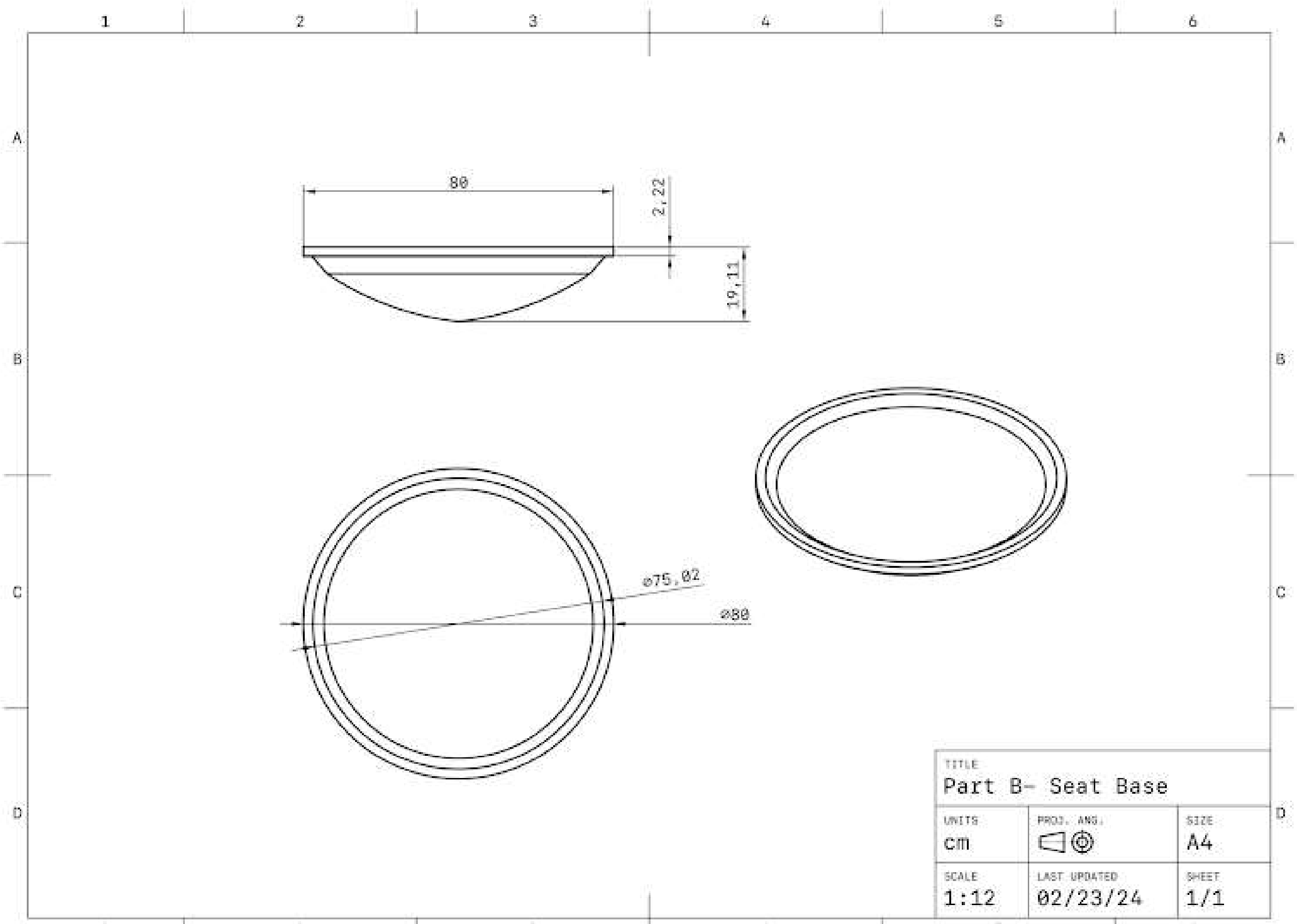

Exploded Drawing
Q Nails
R Nuts
A
Part A is placed on Part B
B
C
Part C is attached to Part B with adhesive
Part C is attached to Part D with nails
Part E is attached to Part B with adhesive
D
E
F
G
Part G is attached to Part E with adhesive
Part F & I are attached to Part E with a screw and bolt
H
Part H is attached to Part G with nails
I
J
Part H is attached to Part D & J with adhesive
Part K is attached to Part J with nails
K
Part L is attached to Part K with adhesive
L
Part L is attached to Part E with a screw and bolt
Part M is attached to Part L with adhesive
M
N
Part M is attached to Part E with adhesive
O
P
Part M is attached to Part N & O with nails
Part P is attached to Part E with a nail

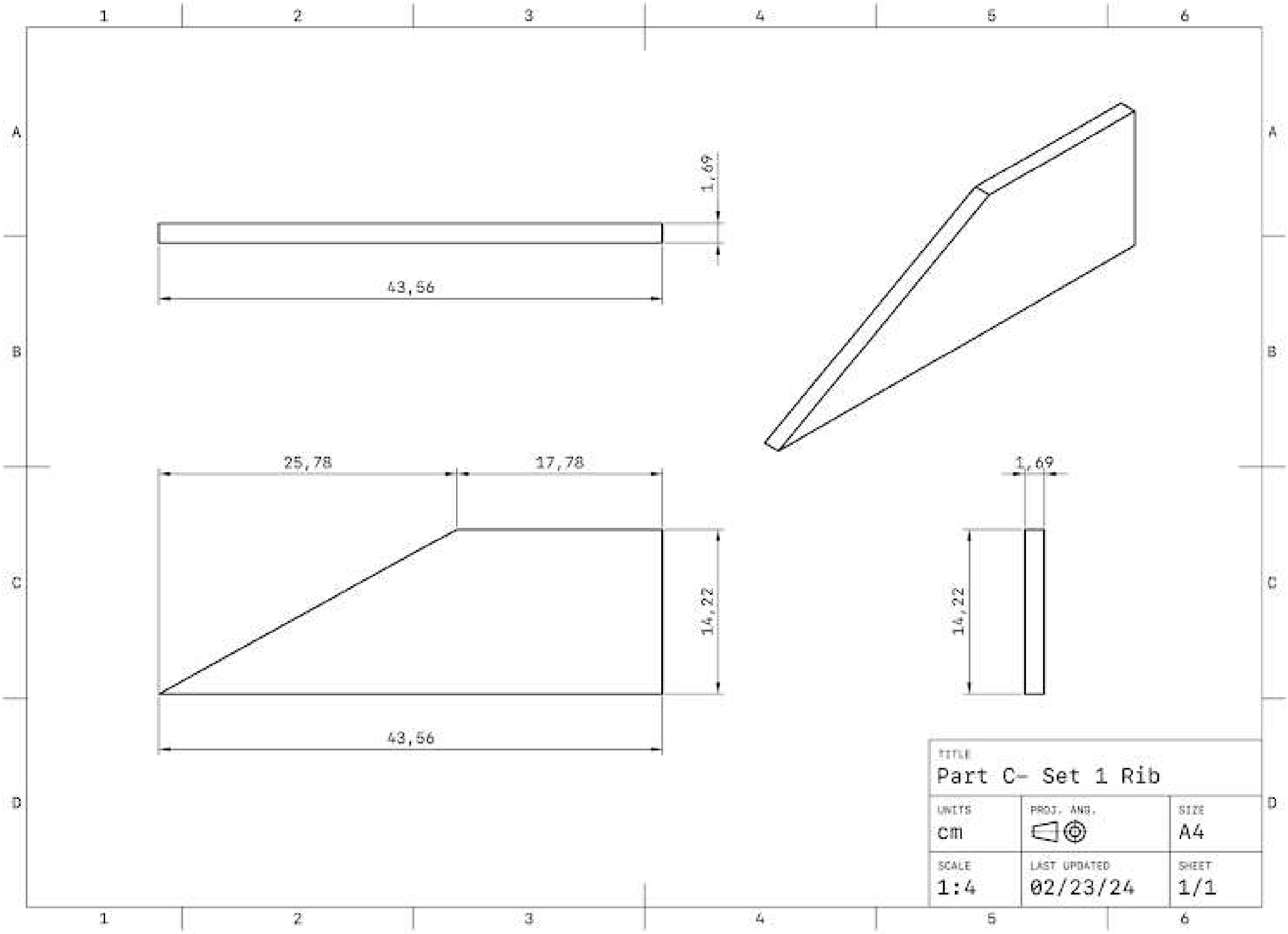
Bill of Materials

Part Letter	Part Name	Material	Dimensions / cm (Flat Wood)			Quantity	Cost
			Length	Width	Thickness		
A	Cushion	Linen Fabric	52.7 8.62	52.7 8.62	200 gsm	2	₹810
		Polyester Fiberfill	52.7	52.7	8.62	1	₹100
		Sewing thread	1 roll/ 100m	0.032	0.032	1	₹80
B	Seat Base	Pine Wood	80.0	80.0	2.22	1	₹108
C	Set 1 Ribs (0.03 cft x 8)	Pine Wood	31.5	15.9	1.68	8	₹168
D	Exterior Frustum 1 (0.53 cft)	Pine Wood	252.2	27.0	2.22	1	₹371
E	Steel Pole	Stainless Steel	49.0	4.85	4.85	1	₹1200
F	Circle piece 1 (0.13 cft)	Pine Wood	40.0	40.0	2.22	1	₹91
G	Steel Spokes	Stainless Steel	17.8	0.533	0.533	8	₹80
H	Hollow Cylinder (0.1 cft)	Pine Wood	136.0	9.78	2.22	1	₹70
I	Circle Piece 2 (0.13 cft)	Pine Wood	40.0	40.0	2.22	1	₹91
J	Exterior Frustum 2 (0.71 cft)	Pine Wood	301.0	30.0	2.22	1	₹497
K	Set 2 Ribs (0.04 cft x 8)	Pine Wood	14.22	43.5	1.68	8	₹224
L	Circle Piece 3 (0.72 cft)	Pine Wood	96.0	96.0	2.22	1	₹504
M	Set 3 Ribs (0.05 cft x 8)	Pine Wood	43.3	19.0	1.68	8	₹280
N	Exterior Frustum 3 (0.71 cft)	Pine Wood	301.59	29.93	2.22	1	₹497
O	Exterior Frustum 4 (0.16 cft)	Pine Wood	122.52	16.32	2.22	1	₹112
P	Circle Base (0.01 cft)	Pine Wood	11.55	11.55	2.22	1	₹7
Q	Nails	Stainless Steel	5.20	0.30	0.30	40	₹50
R	Nuts	Stainless Steel	2.0	5.0	5.0	3	₹260

C2: Orthographic drawings



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1,69
14,22
43,56
TITLE
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SCALE
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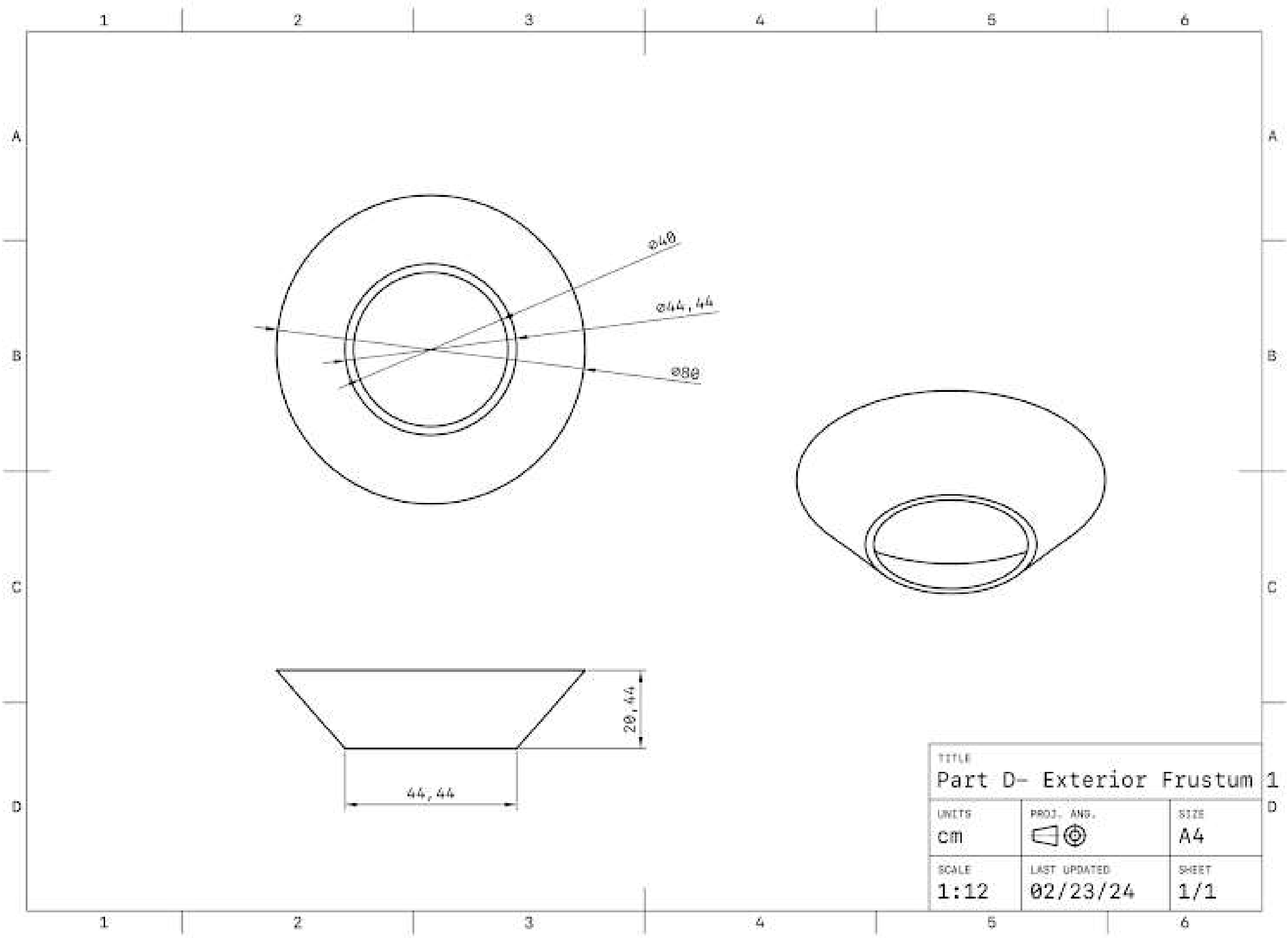
LAST UPDATED
02/23/24
SHEET
1/1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⌀80
20,44
44,44
TITLE
Part D- Exterior Frustum 1
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SIZE
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SCALE
1:12
LAST UPDATED
02/23/24
SHEET
1/1

TITLE		
Part G– Steel Spokes		
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SCALE	LAST UPDATED	SHEET
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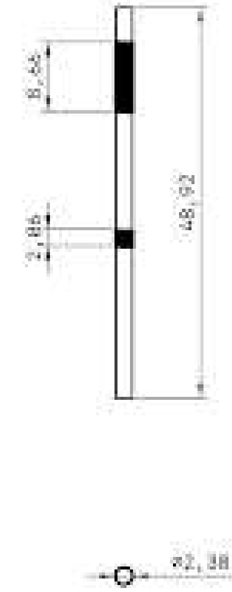

TITLE		
Part E– Steel Pole		
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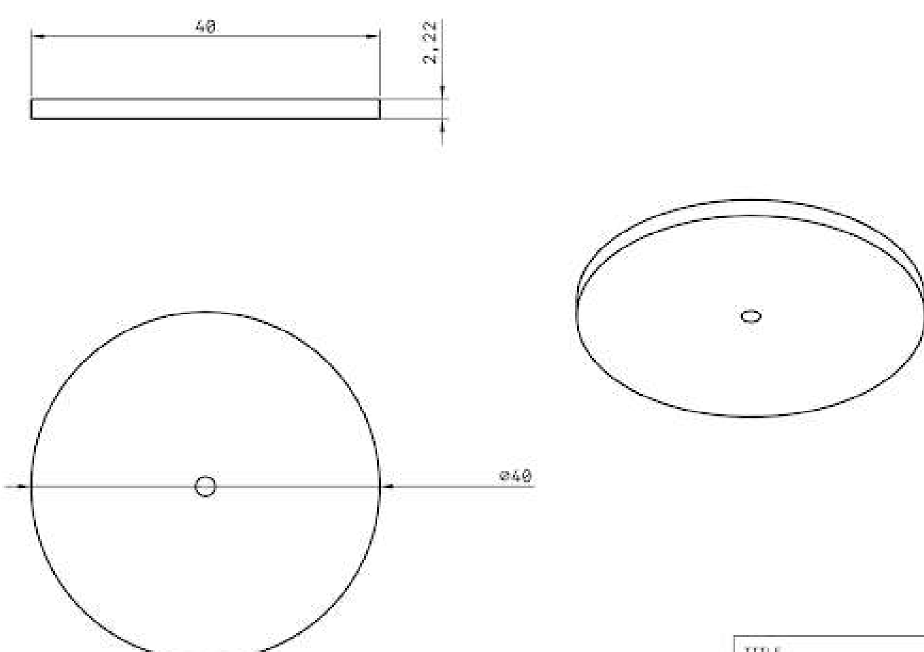

TITLE		
Part F & I– Circle Piece 1 &		
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cm		A4
SCALE	LAST UPDATED	SHEET
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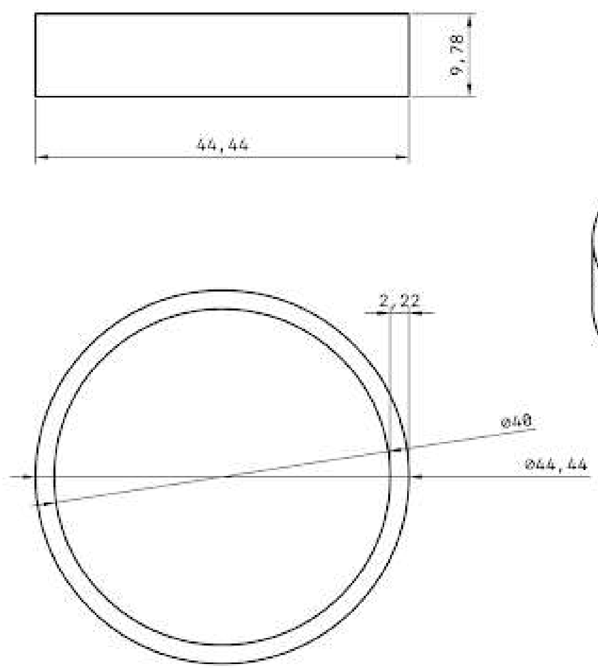
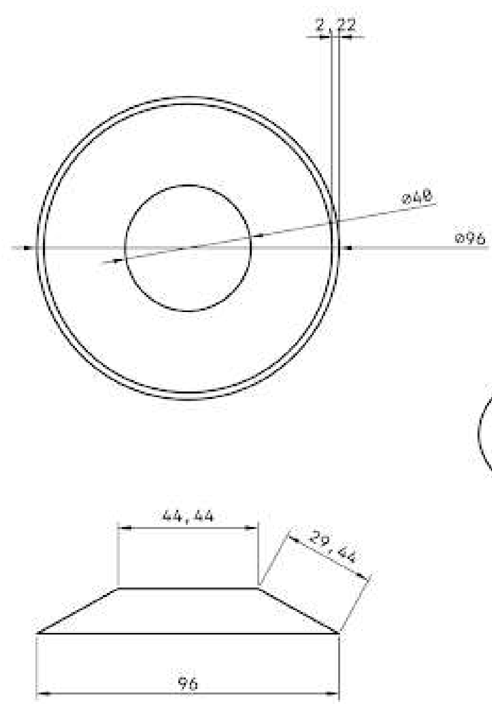


TITLE
Part H- Hollow Cylinder

UNITS	PROJ. ANG.	SIZE
cm		A4
SCALE	LAST UPDATED	SHEET
1:6	02/23/24	1/1



TITLE
Part J- Exterior Frustum 2

UNITS	PROJ. ANG.	SIZE
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SCALE	LAST UPDATED	SHEET
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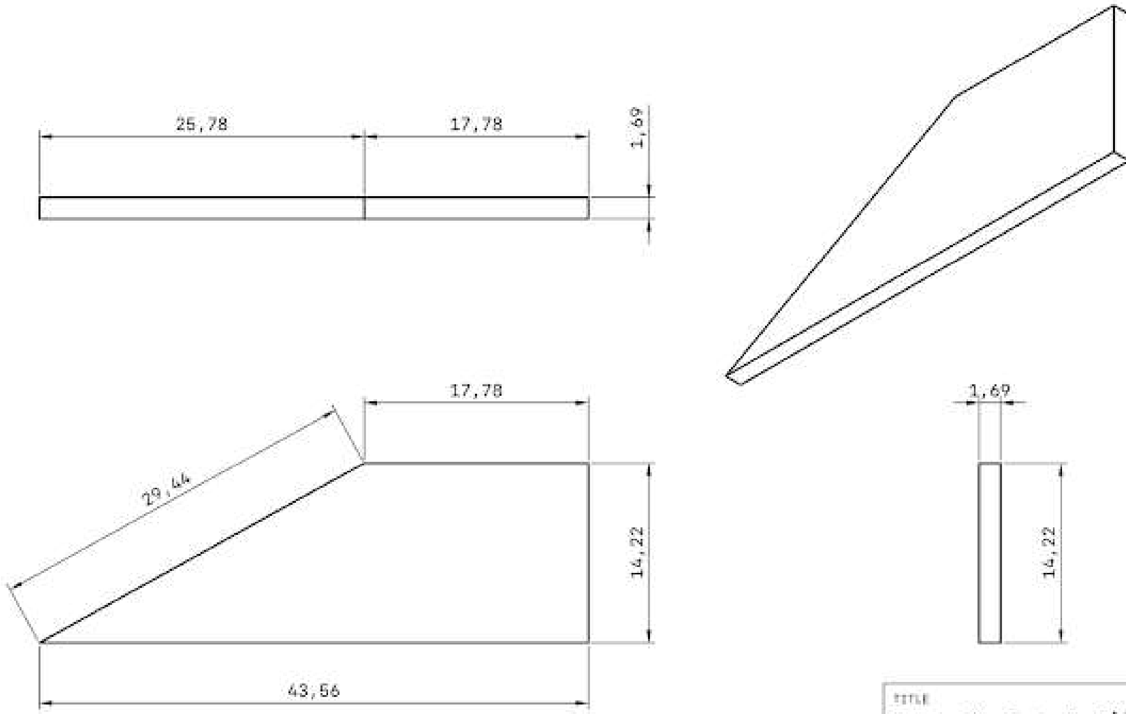

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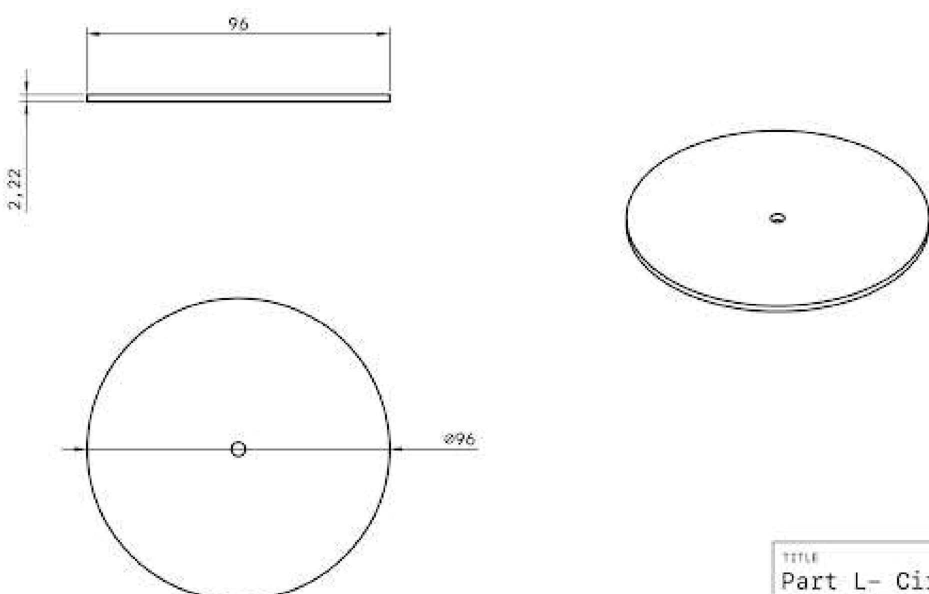

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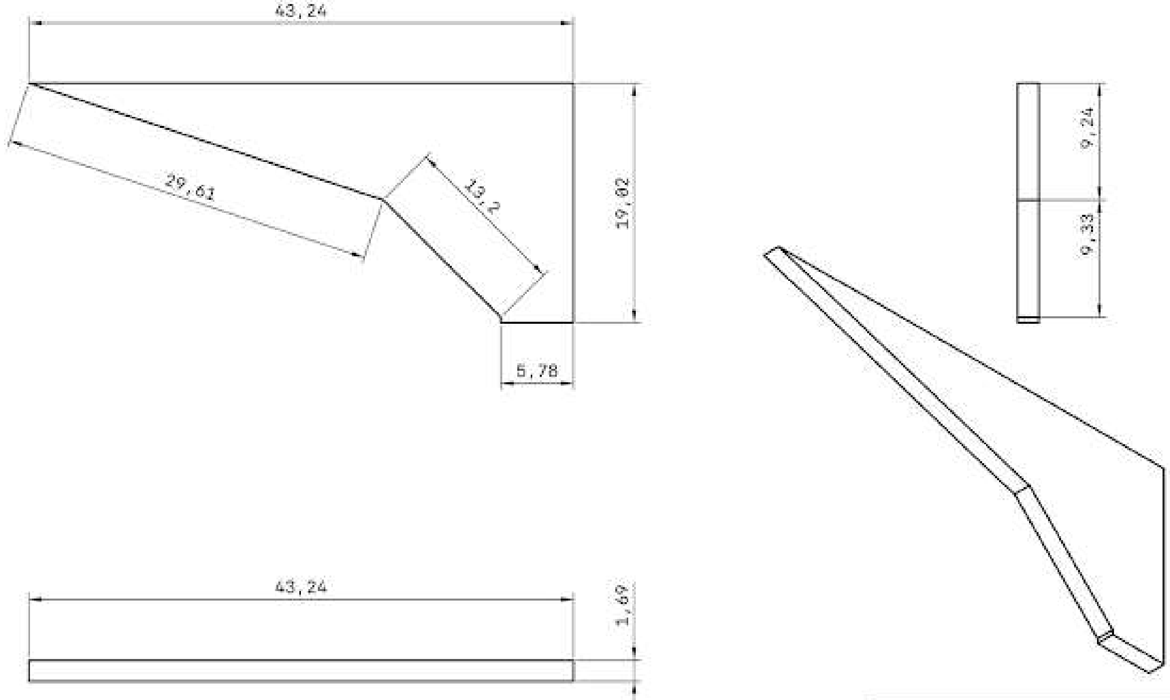

TITLE		
Part M- Set 3 Rib		
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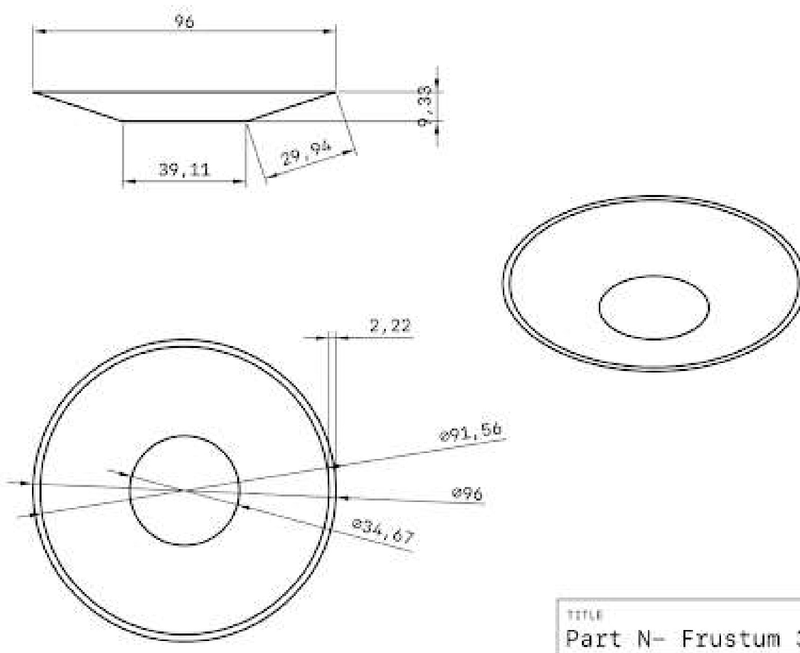

TITLE		
Part N- Frustum 3		
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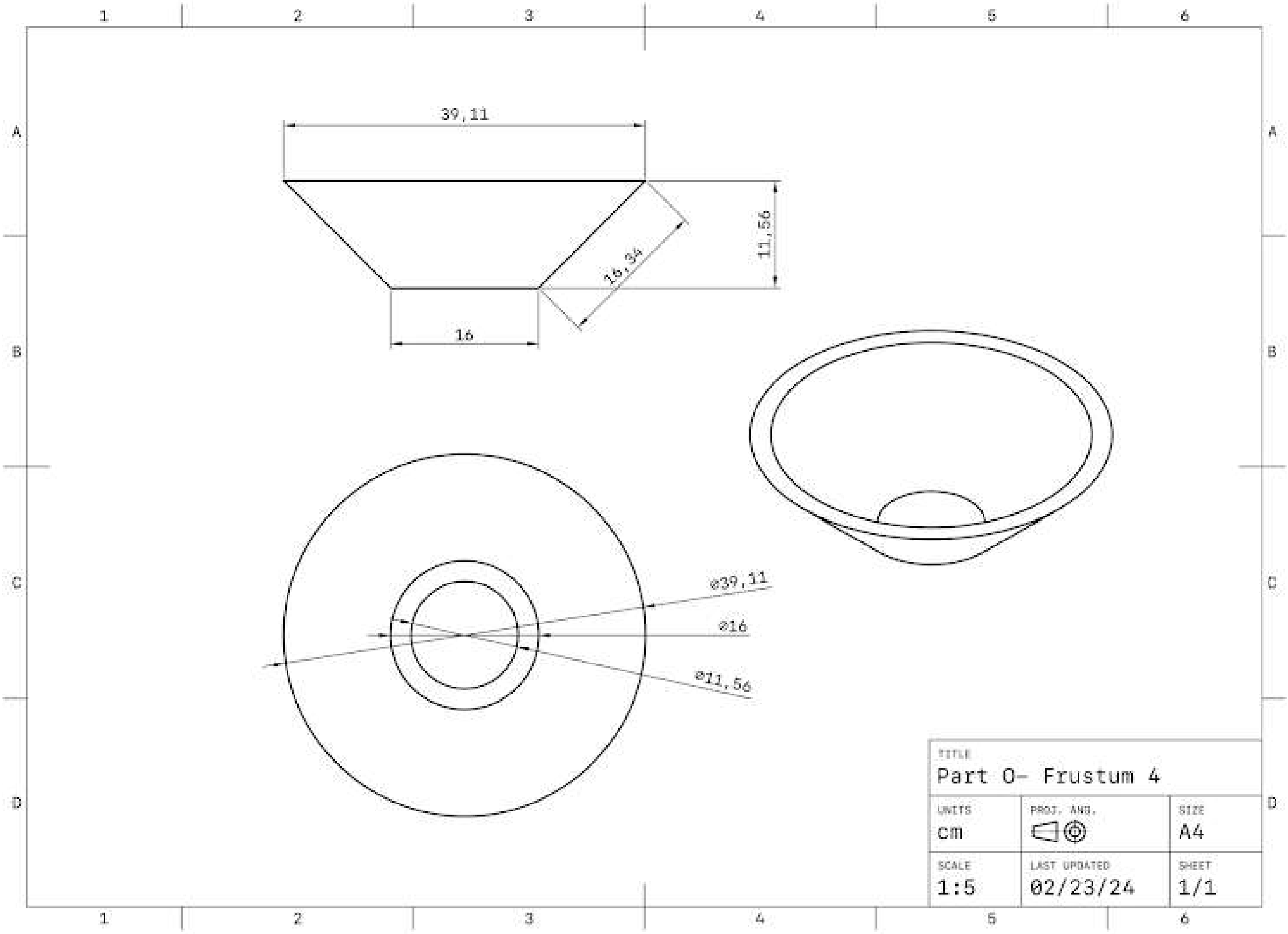
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⌀16
⌀11,56
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Part O- Frustum 4
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PROJ. ANG.
SIZE
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SCALE
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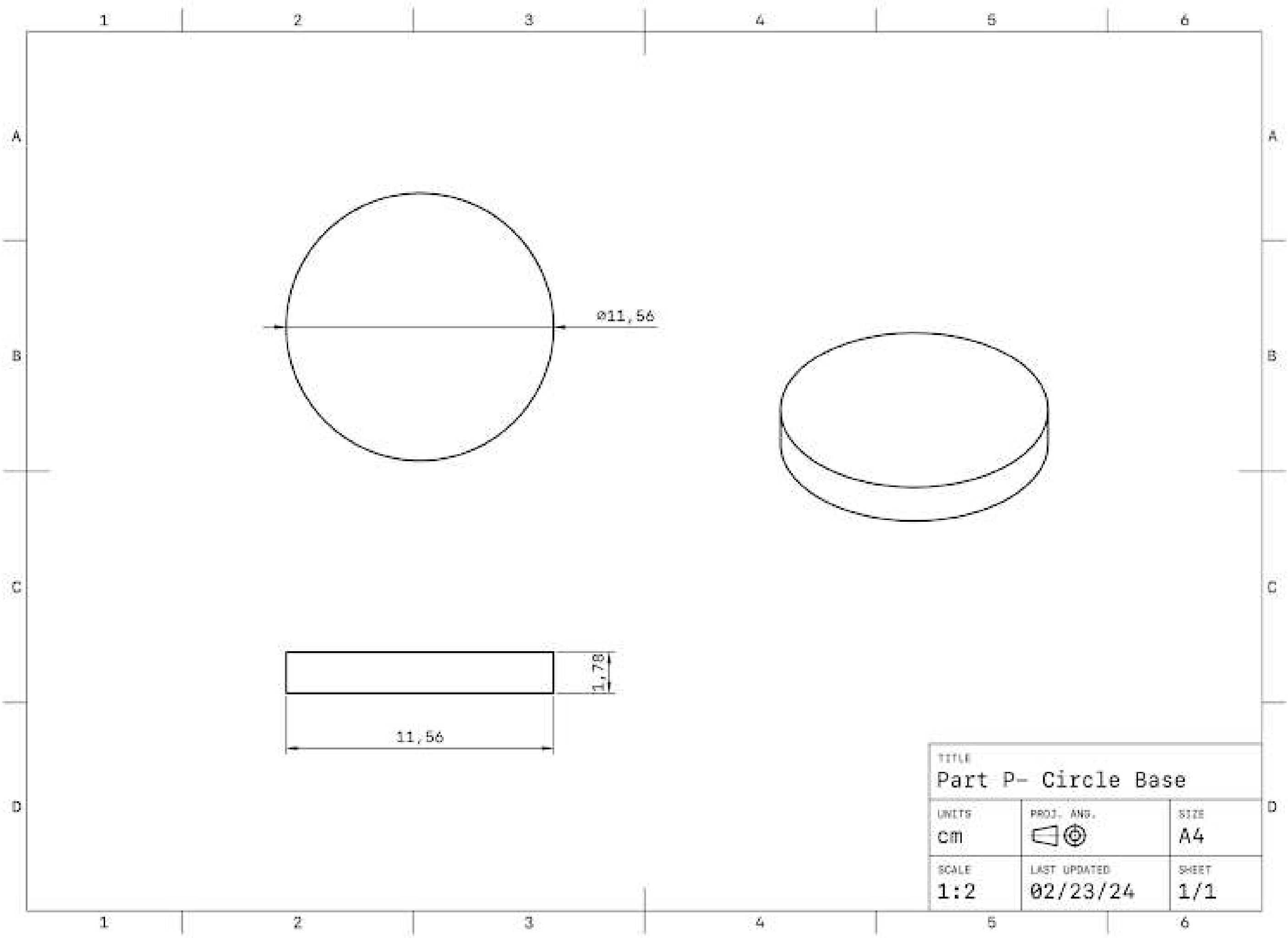
LAST UPDATED
02/23/24
SHEET
1/1


⌀11,56
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11,56
TITLE
Part P- Circle Base
UNITS
cm
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SIZE
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SCALE
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LAST UPDATED
02/23/24
SHEET
1/1

Construction Plan

No.	Processes	Equipment	Schedule	Quality Control	Risk Assessment	Mitigation Strategy
#1	Create digital drawings for all pieces.	2D Design	1 hr	Ensure the dimensions are correct	-	-
#2	Transfer files and laser cut the pieces	USB CNC Router	2 hr	Ensure laser is at the correct height Cleaning lens using acetone.	Harmful fumes, Scattered debris, Fume extractor Ventilator	Fume extractor, Ventilation Safety Goggles, Apron
#3	Make Manual Adjustments	Band Saw, Table Saw, Ruler	1 hr	Ensure precise clean cut according to markings.	Correct hand Placement. (Sharp blade)	Gloves, Safety Goggles, Apron
#4	File and sand the sharp edges	Chisel, File, Sandpaper, Orbital Sander	1 hr	Ensure shaving is according to marking.	Airborne dust and finger abrasion.	Gloves, Safety Goggles, Apron
#5	Paint the wood pieces and apply primer	Latex Paint, Primer Brush	1 hr 1 hr 2 hr	Apply multiple coats and dry out.	Don't make direct contact.	Gloves, Safety Goggles, Apron
#6	Mark the Linen Fabric	Marker, Ruler, Angle Ruler,	30 min	Ensure the dimensions are precise using ruler.	-	-
#7	Cut out the Linen Fabric pieces	Fabric Scissors, Tailors Chalk, Measuring Tape, Rotary Cutter	30 min	Ensure cutting is according to the correct markings.	Correct hand placement (Sharp Blade)	-
#8	Sew the pieces together	Needles, Thread, Sewing, Machine,	30 min	Ensure the sewing is recording to markings.	Correct hand placement (Sharp needle)	-
#9	Fill the pieces with Polyester Fiberfill	-	10 min	Ensure the Cushion is filled solid.	-	-
#10	Mark out the Steel Spokes and Pole	Marker, Angle Ruler, Ruler,	10 min	Ensure the dimensions are precise.	-	-

Construction Plan						
No.	Processes	Equipment	Schedule	Quality Control	Risk Assessment	Safety precaution
#11	Cut out the Steel Spokes and Pole	Angle Grinder, Hacksaw, Belt Sander,	20 min	Ensure the dimensions are precise.	Flying Debris, Correct hand placement, Sharp blade	Gloves, Safety Goggles, Apron
#12	Drill the holes into the pieces	Power Drill,	10 minutes	Ensure holes are consistent.	Correct hand placement	Gloves, Safety Goggles, Apron
#13	Create Ridges into the Steel Pole	Lathes	10 minutes	Ensure the dimensions are precise.	Correct hand placement	Gloves, Safety Goggles, Apron
#14	Carve out placeholder holes for Part B, C, F, G, I, K, L, M, P on Part E (Steel Pole)	Lathes, Power Drill, Belt Sander,	30 min	Ensure the dimensions are precise.	Flying Debris, Correct hand placement, Abrasion	Gloves, Safety Goggles, Apron
#15	Slide through a Steel Pole through the middle of the pieces.	Hand Work	5 min	Ensure the handwork is precise.	Sharp edges	Gloves, Safety Goggles, Apron
#16	Attach Part B, C, F, G, I, K, L, M, P and the Nuts to Part E (Steel Pole)	Adhesive/ Wood glue	1 hr	Spread the glue without bumps.	-	Gloves, Safety Goggles, Apron
#17	Attach Part B, D, H, J, M, N, O, P together	Adhesive/ Wood glue	30 min	Spread the glue without bumps and attach pieces accurately.	Don’t make direct contact with glue	Gloves, Safety Goggles, Apron
#18	Nail Part B, D, H, J, M, N, O, P together	Nails, Hammer,	20 min	Hammer nails with consistency.	Sharp nails,	Gloves, Safety Goggles, Apron

CRITERIA D1 Justify a testing strategy to measure the success of the prototype

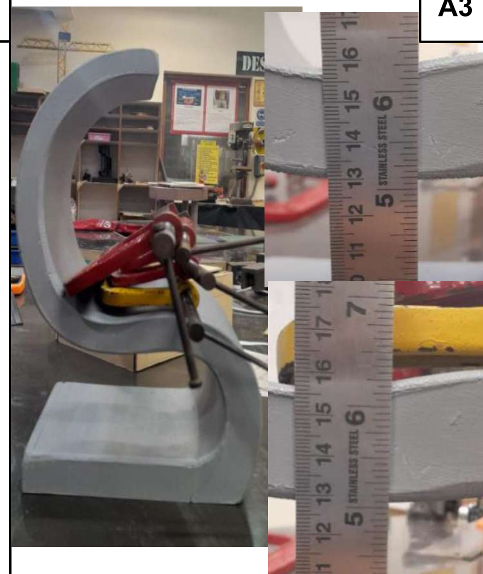
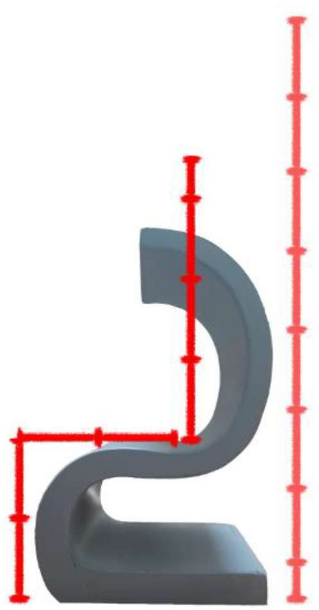
D1 Word Count = 199

Priority	Testing Technique	Justification	Types of data generated
1	**User Interview/Observation:** Assess the anthropometrics of the chair for functionality and comfort. *Specifications: 2.0*	To ensure the chair can accompany the 95th percentile of body measurements with ease.	Quantitative (Anthropometric data)
2	**Performance test:** Testing load bearing capacity using weights in increments to measure the deformation/deflection in chair frame. *Specifications: 1.3, 7.1, 7.2*	Through this test we can determine structural integrity, load bearing capacity, modulus of resistance to ensure there is no compromise in user safety due to fractures.	Quantitative (Weight threshold)
3	**Performance test:** Evaluate the stability, rocking, and tipping under dynamic load through testing rig to simulate movement. *Specifications: 1.3, 7.1, 7.2*	To determine reliability and ensure the chair won't sway or move while the user is in motion.	Qualitative (Observation of stability)
4	**User research (Usability Evaluation):** Interview with Expert in the field. *Specifications: 1.4, 2.0, 7.3, 7.1, 7.2*	Gathering Information on the ergonomics, comfort, finishing, and user experience. Identifying areas for improvement	Qualitative (Expert Feedback)
5	**Performance test:** Measure the pressure points of chair through technology. *Specifications: 1.3, 7.1, 7.2*	Use pressure mapping technology to analyze weight distribution levels for durability flaws and reduce risk of user safety due to fracture.	Quantitative (Pressure mapping data)
6	**Performance test:** Evaluate ease of transport, resistance to abrasion through Drag test. *Specifications: 10.1*	Test susceptibility to abrasion during long term use, determining long-term durability.	Quantitative (Damage severity)
7	**Fit and Finish:** Inspect the prototype and gain expert feedback for the quality of finish, fit, attention to detail, and attention to detail. *Specifications: 7.3*	This is crucial for aesthetic appeal and picks out any possible improvements such as smooth surfaces, edges, consistency of gaps, attention to detail and general craftsmanship.	Qualitative (Expert Feedback)

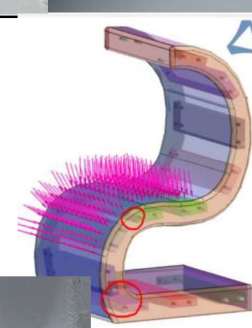
Images of chair prototype

CRITERIA D2 **Evaluate the success of the prototype against the design specification** **D2 Word Count = 698**

Sufficiently met: **Adequately met:** **Not met:**

Success Parameter	**Explanation, Results and Ratings**	**Evidence of Testing**	**Specifications from A3**
Performance test: Testing load bearing capacity using weights in increments at different locations to measure deformation in chair frame and rocking/tipping point	Using a load of approximately 4kg's (three 500g and one 1kg clamp) placed on the seat of the chair, I observed the chair was easily able to withstand the weight but did experience deformation. By measuring the original height and calculating the difference between the height with load, the seat dipped around .3 cm. This was due to the cantilever design whereas it indicated noticeable structural weakness from 4 kg onwards. While this weight is sufficient, strengthening the bottom structural frame of a chair is crucial to prevent fracture.		1.3 7.1 7.2
Performance test: Evaluate the stability, rocking, and tipping under dynamic load through testing rig to simulate movement.	The simulation attempts to recreate the motion of sitting back which applies force on the backrest, which could potentially tilt the chair backwards. Thus, the testing rig composes of an exercise band which is wrapped around the backrest of the chair attached to a door or weights. Through multiple tests, immediate and pulling of the bands at a fast rate caused the chair to tip over regardless of the positioning of the backrest. However, at a slow and general speed the chair was dragged across the surface. When the band was attached higher up on the backseat, there was a significant increase in tipping. Through this I inferred that the faster the dynamic load increases, the more stability required. Whereas, the lower on the backseat the weight is applied, the lower the chances for the seat to tip. This is largely because the base of the chair is too short.	 	1.3 7.1 7.2
User Interview/ Observation: Assess the anthropometrics of the chair for functionality and comfort.	By taking an image of the chair and comparing it with the corresponding proportions of the average human measurements, my results showed that the torso would on average be much taller than the chair indicated. Due to which, they would be unable to sit in the chair, or would have to duck in an awkward position decreasing the comfort largely. These measurements were in accordance to legs that would reach the floor. **Proportions:** *Average Human: 7.5 heads* *Torso & Head: 3.5 heads* *Legs: 4 heads*		2.0

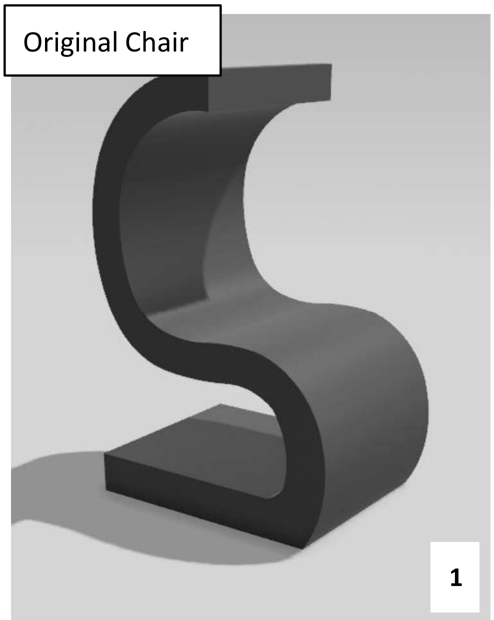
CRITERIA D2 **Evaluate the success of the prototype against the design specification** **D2 Word Count = 698**

Success Parameter	Explanation, Results and Ratings	Evidence of Testing	Specifications
Performance test: Evaluate ease of transport, resistance to abrasion through Drag test.	By assessing the friction accumulated through 3 weeks after the chair was created, we notice that the bottom of the chair and paint job is significantly damaged due to abrasion. While the chair is a prototype created using foam board, the method of transporting the chair is still dragging it. Through which I understand that it requires legs.	 	10.1
User research (Usability Evaluation): Interview with Expert in the field.	**Interviewee:** Mr Deepak Singh **Chair Structure:** 6/10, **Finishing:** 8/10, **Aesthetics:** 8/10 **Ergonomics:** Proportionally, the height of a torso is too short compared to the legs. Moreso, the curved shape of the backrest forces users to duck head forward reducing comfort. **Improvements:** The sitting surface can be further curved and armrests can be attached so users can situate themselves in a more secure and comfortable seat. As a cantilever chair, to make it more structurally sound the bottom area of the chair can be utilized for storage such as drawers for support. Tiny legs can be added to allow easy cleaning when the floor is cleaned using bottom which could potentially soak and harm the chair material. Lastly, suggestions were provided for Fit and Finish which is included in the row below.		1.4 2.0 7.3 7.1 7.2
Fit and Finish: Inspect the prototype and gain expert feedback for the quality of finish, fit, attention to detail, and attention to detail.	By analysis and expert interview, the chair had an adequate finish utilizing beveled edges to avoid sharp edges that could hinder the users safety and smooth surfaces. The overall rating for the finish was 8/10 however the chair was slightly misaligned and the expert pointed out how the panels in the chair frame could have been more effectively placed. Where the expert mentioned how I could have analyzed the chair first and divided them to optimize load distribution, for the base this meant lowering the panels.	  	7.3
Performance test: Measure the pressure points of chair through technology.	By using the application, SIMSOLID, and FEA technology, I was able to upload my design and simulate the various loads on the chair due to which the chair may fracture. Whereas, the software would then calculate the material, density, young's modulus, and force applied (95th percentile of a person sitting) which was 250 pounds. While there was no fracture, the chair did experience deformation. The pressure points were then identified, where one was through the FEA, and the other was through a practical test previously which led to the board snapping.	 	1.3 7.1 7.2

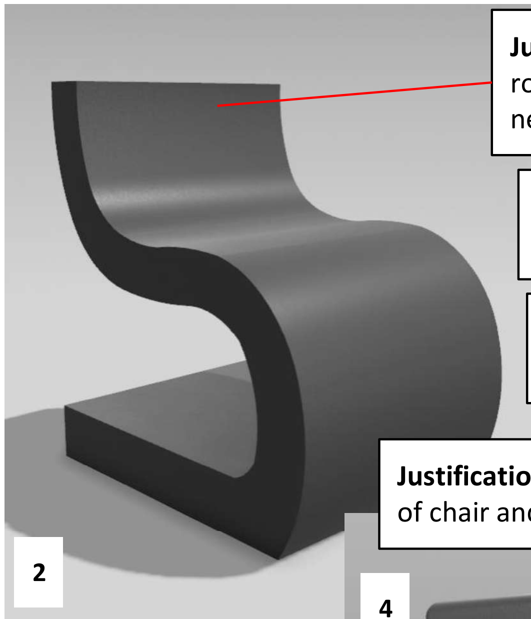
CRITERIA D3 Demonstrate how the prototype could be improved

Arill Shukla D3 Word Count = 0

Original Chair



1



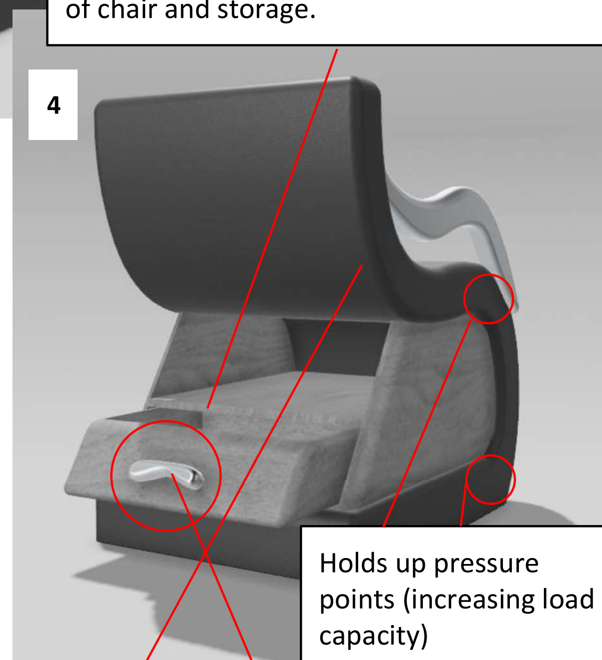
2

Justification: By removing the roof, the torso height is negligible.

Roof of the chair no longer takes as much space.

Requires less material for construction.

Justification: Drawers to prevent collapse of chair and storage.



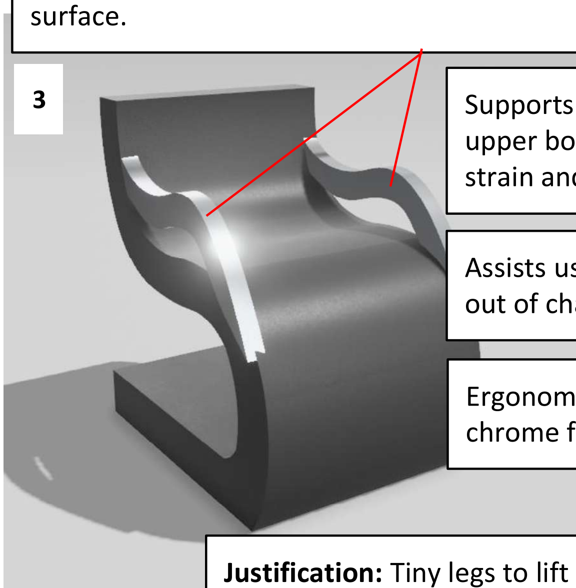
4

Holds up pressure points (increasing load capacity)

Further bevelling to prevent scratches, increasing safety.

Handle for easy access and maximizing storage area.

Justification: Armrests prevent user falling off and provide arms surface.



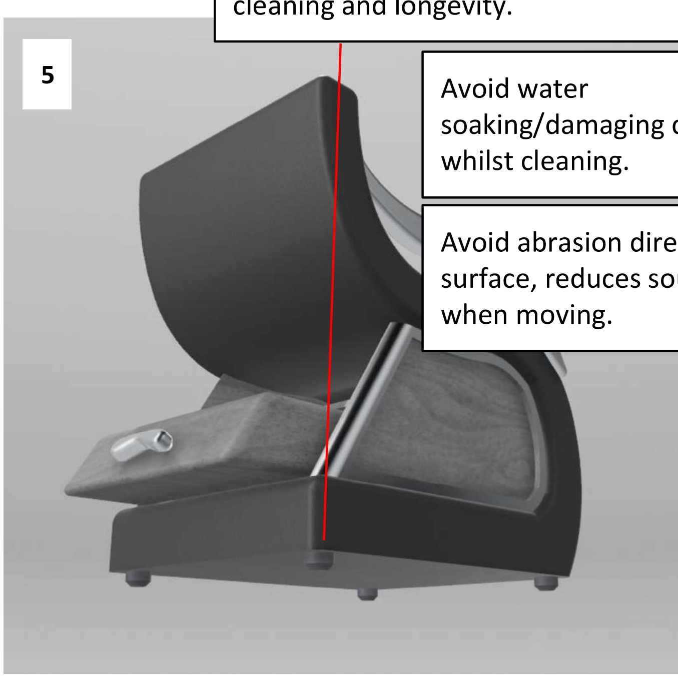
3

Supports arms, shoulder, upper body to reduce strain and fatigue.

Assists user to get in and out of chair easier.

Ergonomic and polished chrome for aesthetic.

Justification: Tiny legs to lift chair- easier cleaning and longevity.

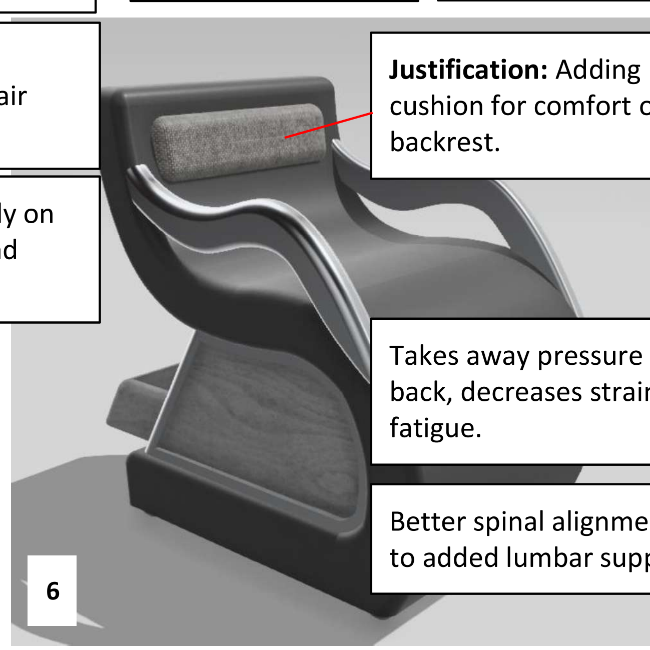


5

Avoid water soaking/damaging chair whilst cleaning.

Avoid abrasion directly on surface, reduces sound when moving.

Justification: Adding cushion for comfort on backrest.



6

Takes away pressure from back, decreases strain and fatigue.

Better spinal alignment due to added lumbar support.